

Create Eureka Discovery Server and register microservices

1. eureka-discovery-server

pom.xml

xml

```
<?xml version="1.0" encoding="UTF-8"?>

<project xmlns="http://maven.apache.org/POM/4.0.0"
  xmlns:xsi="http://www.w3.org/2001/XMLSchema-instance"
  xsi:schemaLocation="http://maven.apache.org/POM/4.0.0
    https://maven.apache.org/xsd/maven-4.0.0.xsd">

  <modelVersion>4.0.0</modelVersion>

  <parent>

    <groupId>org.springframework.boot</groupId>

    <artifactId>spring-boot-starter-parent</artifactId>

    <version>4.1.0</version>

    <relativePath/>

  </parent>

  <groupId>com.cognizant</groupId>

  <artifactId>eureka-discovery-server</artifactId>

  <version>0.0.1-SNAPSHOT</version>

  <properties>

    <java.version>17</java.version>

    <spring-cloud.version>2025.1.2</spring-cloud.version>

  </properties>

  <dependencies>

    <dependency>

      <groupId>org.springframework.cloud</groupId>

      <artifactId>spring-cloud-starter-netflix-eureka-server</artifactId>

    </dependency>
```

```

<dependency>
  <groupId>org.springframework.boot</groupId>
  <artifactId>spring-boot-starter-test</artifactId>
  <scope>test</scope>
</dependency>
</dependencies>
<dependencyManagement>
  <dependencies>
    <dependency>
      <groupId>org.springframework.cloud</groupId>
      <artifactId>spring-cloud-dependencies</artifactId>
      <version>${spring-cloud.version}</version>
      <type>pom</type>
      <scope>import</scope>
    </dependency>
  </dependencies>
</dependencyManagement>
<build>
  <plugins>
    <plugin>
      <groupId>org.springframework.boot</groupId>
      <artifactId>spring-boot-maven-plugin</artifactId>
    </plugin>
  </plugins>
</build>
</project>

```

src/main/java/com/cognizant/eureka_discovery_server/EurekaDiscoveryServerApplication.java

```

java

package com.cognizant.eureka_discovery_server;


import org.springframework.boot.SpringApplication;
import org.springframework.boot.autoconfigure.SpringBootApplication;
import org.springframework.cloud.netflix.eureka.server.EnableEurekaServer;


@SpringBootApplication
@EnableEurekaServer

public class EurekaDiscoveryServerApplication {

    public static void main(String[] args) {

        SpringApplication.run(EurekaDiscoveryServerApplication.class, args);

    }

}

```

src/main/resources/application.properties

```

properties

server.port=8761

eureka.client.register-with-eureka=false
eureka.client.fetch-registry=false

logging.level.com.netflix.eureka=OFF
logging.level.com.netflix.discovery=OFF

```

2. account

pom.xml (dependencies section — keep the rest of the pom same shape as above)

```

xml

<dependencies>

    <dependency>

        <groupId>org.springframework.boot</groupId>

```

```

        <artifactId>spring-boot-starter-web</artifactId>
    </dependency>
    <dependency>
        <groupId>org.springframework.boot</groupId>
        <artifactId>spring-boot-devtools</artifactId>
        <scope>runtime</scope>
        <optional>true</optional>
    </dependency>
    <dependency>
        <groupId>org.springframework.cloud</groupId>
        <artifactId>spring-cloud-starter-netflix-eureka-client</artifactId>
    </dependency>
    <dependency>
        <groupId>org.springframework.boot</groupId>
        <artifactId>spring-boot-starter-test</artifactId>
        <scope>test</scope>
    </dependency>
</dependencies>

```

Artifact account, so the <artifactId> at top and dependencyManagement block are the same pattern as the eureka server's pom — just swap the artifact name and dependencies.

src/main/java/com/cognizant/account/AccountApplication.java

```

java

package com.cognizant.account;

import org.springframework.boot.SpringApplication;
import org.springframework.boot.autoconfigure.SpringBootApplication;
import org.springframework.cloud.client.discovery.EnableDiscoveryClient;

```

```
@SpringBootApplication
@EnableDiscoveryClient

public class AccountApplication {

    public static void main(String[] args) {

        SpringApplication.run(AccountApplication.class, args);

    }

}
```

src/main/resources/application.properties

```
properties

spring.application.name=account-service

server.port=8081

eureka.client.service-url.defaultZone=http://localhost:8761/eureka/
```

3. loan

Same as account, just:

src/main/java/com/cognizant/loan/LoanApplication.java

```
java

package com.cognizant.loan;

import org.springframework.boot.SpringApplication;
import org.springframework.boot.autoconfigure.SpringBootApplication;
import org.springframework.cloud.client.discovery.EnableDiscoveryClient;

@SpringBootApplication
@EnableDiscoveryClient

public class LoanApplication {

    public static void main(String[] args) {

        SpringApplication.run(LoanApplication.class, args);

    }

}
```

```
}  
}
```

src/main/resources/application.properties

properties

spring.application.name=loan-service

server.port=8082

eureka.client.service-url.defaultZone=http://localhost:8761/eureka/

4. greet-service

pom.xml (dependencies section)

xml

```
<dependencies>
```

```
  <dependency>
```

```
    <groupId>org.springframework.boot</groupId>
```

```
    <artifactId>spring-boot-starter-web</artifactId>
```

```
  </dependency>
```

```
  <dependency>
```

```
    <groupId>org.springframework.cloud</groupId>
```

```
    <artifactId>spring-cloud-starter-netflix-eureka-client</artifactId>
```

```
  </dependency>
```

```
  <dependency>
```

```
    <groupId>org.springframework.boot</groupId>
```

```
    <artifactId>spring-boot-starter-test</artifactId>
```

```
    <scope>test</scope>
```

```
  </dependency>
```

```
</dependencies>
```

src/main/java/com/cognizant/greet_service/GreetServiceApplication.java

java

```
package com.cognizant.greet_service;
```

```
import org.springframework.boot.SpringApplication;
```

```
import org.springframework.boot.autoconfigure.SpringBootApplication;
```

```
@SpringBootApplication
```

```
public class GreetServiceApplication {
```

```
    public static void main(String[] args) {
```

```
        SpringApplication.run(GreetServiceApplication.class, args);
```

```
    }
```

```
}
```

```
src/main/java/com/cognizant/greet_service/GreetController.java
```

```
java
```

```
package com.cognizant.greet_service;
```

```
import org.springframework.web.bind.annotation.GetMapping;
```

```
import org.springframework.web.bind.annotation.RestController;
```

```
@RestController
```

```
public class GreetController {
```

```
    @GetMapping("/greet")
```

```
    public String greet() {
```

```
        return "Hello World";
```

```
    }
```

```
}
```

```
src/main/resources/application.properties
```

```
properties
```

spring.application.name=greet-service

server.port=8083

eureka.client.service-url.defaultZone=http://localhost:8761/eureka/

5. api-gateway

pom.xml (matches yours, dependencies section)

xml

```
<dependencies>
```

```
  <dependency>
```

```
    <groupId>org.springframework.cloud</groupId>
```

```
    <artifactId>spring-cloud-starter-gateway-server-webmvc</artifactId>
```

```
  </dependency>
```

```
  <dependency>
```

```
    <groupId>org.springframework.cloud</groupId>
```

```
    <artifactId>spring-cloud-starter-netflix-eureka-client</artifactId>
```

```
  </dependency>
```

```
  <dependency>
```

```
    <groupId>org.springframework.boot</groupId>
```

```
    <artifactId>spring-boot-starter-test</artifactId>
```

```
    <scope>test</scope>
```

```
  </dependency>
```

```
</dependencies>
```

src/main/java/com/cognizant/api_gateway/ApiGatewayApplication.java

java

```
package com.cognizant.api_gateway;
```

```
import org.springframework.boot.SpringApplication;
```

```
import org.springframework.boot.autoconfigure.SpringBootApplication;
```

```
@SpringBootApplication
public class ApiGatewayApplication {
    public static void main(String[] args) {
        SpringApplication.run(ApiGatewayApplication.class, args);
    }
}
```

src/main/java/com/cognizant/api_gateway/LogFilter.java

java

```
package com.cognizant.api_gateway;
```

```
import jakarta.servlet.Filter;
import jakarta.servlet.FilterChain;
import jakarta.servlet.ServletException;
import jakarta.servlet.ServletRequest;
import jakarta.servlet.ServletResponse;
import jakarta.servlet.http.HttpServletRequest;
import org.springframework.stereotype.Component;
```

```
import java.io.IOException;
```

```
@Component
```

```
public class LogFilter implements Filter {
```

```
    @Override
```

```
    public void doFilter(ServletRequest request, ServletResponse response, FilterChain
chain)
```

```
        throws IOException, ServletException {
```

```

HttpServletRequest httpRequest = (HttpServletRequest) request;

System.out.println("Incoming request path: " + httpRequest.getRequestURI());

chain.doFilter(request, response);

}

}

```

src/main/resources/application.properties

properties

spring.application.name=api-gateway

server.port=9090

eureka.client.service-url.defaultZone=http://localhost:8761/eureka/

spring.cloud.gateway.server.webmvc.routes[0].id=greet-route

spring.cloud.gateway.server.webmvc.routes[0].uri=lb://GREET-SERVICE

spring.cloud.gateway.server.webmvc.routes[0].predicates[0]=Path=/greet-service/**

spring.cloud.gateway.server.webmvc.routes[0].filters[0]=RewritePath=/greet-service

The screenshot shows the Spring Eureka dashboard in a web browser. The dashboard is titled "spring Eureka" and has a navigation bar with "HOME" and "LAST 1000 SINCE STARTUP".

System Status

Environment	test	Current time	2026-07-08T18:06:23+0530
Data center	default	Uptime	00:00
		Lease expiration enabled	false
		Renews threshold	1
		Renews (last min)	0

DS Replicas

Instances currently registered with Eureka

Application	Addr	Availability Zones	Status
No instances available			

General Info

Name	Value
total-avail-memory	144mb
num-of-cpus	16
current-memory-usage	66mb (45%)
server-up-time	00:00
registered-replicas	
unavailable-replicas	
available-replicas	

Instance Info

Name	Value
ipAddr	192.168.196.187
status	UP



HOME | LAST 1000 SINCE STARTUP

System Status

Environment

test

Data center

default

Current time

2026-07-08T16:07:34+05:30

Uptime

00:01

Lease expiration enabled

false

Renews threshold

3

Renews (last min)

0

DS Replicas

Instances currently registered with Eureka

Application	Addr	Availability Zones	Status
ACCOUNT-SERVICE	n/a (1)	(1)	UP (1) - localhost:account-service-8081

General Info

Name	Value
total-avail-memory	144mb
num-of-cpus	16
current-memory-usage	84mb (59%)
server-up-time	00:01
registered-replicas	
unavailable-replicas	
available-replicas	

Instance Info

Name	Value
ipAddr	192.168.196.187
status	UP



HOME | LAST 1000 SINCE STARTUP

System Status

Environment

test

Data center

default

Current time

2026-07-08T16:11:02+05:30

Uptime

00:04

Lease expiration enabled

false

Renews threshold

5

Renews (last min)

2

DS Replicas

Instances currently registered with Eureka

Application	Addr	Availability Zones	Status
ACCOUNT-SERVICE	n/a (1)	(1)	UP (1) - localhost:account-service-8081
LOAN-SERVICE	n/a (1)	(1)	UP (1) - localhost:loan-service-8082

General Info

Name	Value
total-avail-memory	144mb
num-of-cpus	16
current-memory-usage	87mb (60%)
server-up-time	00:04
registered-replicas	
unavailable-replicas	
available-replicas	

Instance Info

Name	Value
ipAddr	192.168.196.187
status	UP



HOME | LAST 1000 SINCE STARTUP

System Status

Environment

test

Data center

default

Current time

2026-07-08T16:26:32+05:30

Uptime

00:20

Lease expiration enabled

false

Renews threshold

6

Renews (last min)

4

EMERGENCY! EUREKA MAY BE INCORRECTLY CLAIMING INSTANCES ARE UP WHEN THEY'RE NOT. RENEWALS ARE LESSER THAN THRESHOLD AND HENCE THE INSTANCES ARE NOT BEING EXPIRED JUST TO BE SAFE.

DS Replicas

Instances currently registered with Eureka

Application	Addr	Availability Zones	Status
ACCOUNT-SERVICE	n/a (1)	(1)	UP (1) - localhost:account-service-8081
CREDIT-SERVICE	n/a (1)	(1)	UP (1) - localhost:credit-service-8083
LOAN-SERVICE	n/a (1)	(1)	UP (1) - localhost:loan-service-8082

General Info

Name	Value
total-avail-memory	144mb
num-of-cpus	16
current-memory-usage	96mb (66%)
server-up-time	00:20
registered-replicas	
unavailable-replicas	
available-replicas	

Instance Info

Name	Value
ipAddr	192.168.196.187
status	UP



```
2026-07-08T16:39:49.485+05:30 INFO 21756 --- [api-gateway] [main] b.w.c.s.WebApplicationContextInitializer : Root WebApplicationContext: initialization completed in 1.726 seconds
2026-07-08T16:39:50.937+05:30 INFO 21756 --- [api-gateway] [main] o.s.c.cloud.commons.util.InetUtils : Cannot determine local hostname
2026-07-08T16:39:51.952+05:30 INFO 21756 --- [api-gateway] [main] o.s.c.cloud.commons.util.InetUtils : Cannot determine local hostname
2026-07-08T16:39:52.102+05:30 WARN 21756 --- [api-gateway] [main] org.springframework.cloud.netflix.eureka.client.DiscoveryClientOptionalArgsConfiguration : Eureka HTTP Client uses RestClient.
2026-07-08T16:39:52.114+05:30 INFO 21756 --- [api-gateway] [main] org.springframework.cloud.netflix.eureka.client.DiscoveryClientOptionalArgsConfiguration : Spring Cloud LoadBalancer is currently working with the default implementation of org.springframework.cloud.loadbalancer.core.RoundRobinLoadBalancer
2026-07-08T16:39:52.132+05:30 INFO 21756 --- [api-gateway] [main] o.s.c.n.eureka.InstanceInfoFactory : Setting initial instance status as: STARTING
2026-07-08T16:39:52.138+05:30 INFO 21756 --- [api-gateway] [main] com.netflix.discovery.DiscoveryClient : Initializing Eureka in region us-east-1
2026-07-08T16:39:52.138+05:30 INFO 21756 --- [api-gateway] [main] c.n.d.s.c.aws.ConfigClusterResolver : Resolving eureka endpoints via configuration
2026-07-08T16:39:52.138+05:30 INFO 21756 --- [api-gateway] [main] com.netflix.discovery.DiscoveryClient : Disable delta property : false
2026-07-08T16:39:52.138+05:30 INFO 21756 --- [api-gateway] [main] com.netflix.discovery.DiscoveryClient : Single vip registry refresh property : null
2026-07-08T16:39:52.138+05:30 INFO 21756 --- [api-gateway] [main] com.netflix.discovery.DiscoveryClient : Force full registry fetch : false
2026-07-08T16:39:52.138+05:30 INFO 21756 --- [api-gateway] [main] com.netflix.discovery.DiscoveryClient : Application is null : false
2026-07-08T16:39:52.138+05:30 INFO 21756 --- [api-gateway] [main] com.netflix.discovery.DiscoveryClient : Registered Applications size is zero : true
2026-07-08T16:39:52.138+05:30 INFO 21756 --- [api-gateway] [main] com.netflix.discovery.DiscoveryClient : Application version is -1: true
2026-07-08T16:39:52.138+05:30 INFO 21756 --- [api-gateway] [main] com.netflix.discovery.DiscoveryClient : Getting all instance registry info from the eureka server
2026-07-08T16:39:52.138+05:30 INFO 21756 --- [api-gateway] [main] com.netflix.discovery.DiscoveryClient : The response status is 200
2026-07-08T16:39:52.138+05:30 INFO 21756 --- [api-gateway] [main] com.netflix.discovery.DiscoveryClient : Starting heartbeat executor: renew interval is: 30
2026-07-08T16:39:52.138+05:30 INFO 21756 --- [api-gateway] [main] c.n.d.s.c.aws.ConfigClusterResolver : InstanceInfoReplicator onDemand update allowed rate per 10 seconds
2026-07-08T16:39:52.138+05:30 INFO 21756 --- [api-gateway] [main] com.netflix.discovery.DiscoveryClient : Discovery Client initialized at timestamp 1783508992311 with status 200
2026-07-08T16:39:52.138+05:30 INFO 21756 --- [api-gateway] [main] o.s.c.n.e.s.EurekaServiceRegistry : Registering application API-GATEWAY with eureka with status 200
2026-07-08T16:39:52.138+05:30 INFO 21756 --- [api-gateway] [main] com.netflix.discovery.DiscoveryClient : Saw local status change event StatusChangeEvent [timestamp=1783508992311, currentStatus=STARTING, previousStatus=STARTING]
2026-07-08T16:39:52.138+05:30 INFO 21756 --- [api-gateway] [main] com.netflix.discovery.DiscoveryClient : DiscoveryClient_API-GATEWAY/localhost:api-gateway:9090
2026-07-08T16:39:52.138+05:30 INFO 21756 --- [api-gateway] [main] o.s.boot.tomcat.TomcatWebServer : Tomcat started on port 9090 (http) with context path /
2026-07-08T16:39:52.138+05:30 INFO 21756 --- [api-gateway] [main] o.s.c.n.e.s.EurekaAutoServiceRegistration : Updating port to 9090
2026-07-08T16:39:52.138+05:30 INFO 21756 --- [api-gateway] [main] com.netflix.discovery.DiscoveryClient : DiscoveryClient_API-GATEWAY/localhost:api-gateway:9090
2026-07-08T16:39:52.138+05:30 INFO 21756 --- [api-gateway] [main] o.s.c.cloud.commons.util.InetUtils : Cannot determine local hostname
2026-07-08T16:39:52.138+05:30 INFO 21756 --- [api-gateway] [main] o.s.c.cloud.commons.util.InetUtils : Cannot determine local hostname
2026-07-08T16:39:53.336+05:30 INFO 21756 --- [api-gateway] [main] c.c.api_gateway.ApiGatewayApplication : Started ApiGatewayApplication in 6.726 seconds (process id 21756)
```